CNS MSE - 2 Solution

VIT Vidyalankar Institute of Technology
(Accredited A+ by NAAC)
(Autonomous Institute Affiliated to University of Mumbai)

Mid Semester Examination

Branch	Date	Sem.	Roll No. / Exam Seat No.	Subject	Student's Signature	Junior Supervisor's Name and Sign

Question No.	A	B	C	D	E	F	G	H	Total	Total out of (20 /30 / 40)
1										
2										
3										
4										

Examiners Signature	Student's Sign (After receiving the assessed answer sheet)

Q1. a) Network Access Control (NAC) is a Security framework that manages and controls access to a network by users and devices, ensuring that only authorizers users get access. NAC Solutions provide a Combination of authentication, authorization & monitoring which helps to prevent Unauthorized access.

NAC works on the principle of enforcing strict Security measures before & during new access of it involves.

1) Authentication
2) Authorization
3) Compliance checks.
4) Monitoring & enforcement.

The key NAC technologies are :-

1) 802.1x
2) Pre admission Control
3) Post " Control

NAC can be implemented as:
1. Assessing the Current n/w infrastructure
2. Defining Security Policies
3. Selecting a NAC Solution
4. Device Authentication
5. Compliance check
6. Deployment & integration
7. Monitoring & response
8. User training awareness
9. Continuous review & improvement

b)) SSH is a cryptographic network Protocol used to Securely access & manage n/w devices & Systems remotely. It is widely used for Secure data Communication, remote Command line login & other network services.

The need for SSH in the n/w:

1. Secure remote access: SSH enables administrators & users to Securely access remote Servers, n/w devices or any other Systems over a n/w.
2. Encrypted data transmission: SSH encrypts all traffic between client & Server, Protecting Sensitive data like login Credentials etc.
3. Authentication & Authorization: SSH use Strong authentication mechanism such as Pwd based login, Public Key authentication & 2FA ensuring only authorized users can access the n/w resources

Two application:
1. Remote Server Management
2. Secure File transfer (SCP/SFTP).

9) S1 : Choose two Prime Number P & q

P = 3

q = 11

S2 : Compute n

n = P * q

= 3 * 11

= 33

S3 : Compute $\phi(n)$

$\phi(n) = (P-1) * (q-1)$

$= 2 \times 10 = 20$

S4 : choose e ∴ e = 3

S5 : calculate d.

$d * e = 1 \bmod 20$

∴ d = 7

S6 : ∴ Public key (e, n) = (3, 33)

Private key (d, n) = (7, 33)

encryption : $C = M^e \bmod n$

M = 8, e = 3, n = 33

$C = 8^3 \bmod 33 = 512 \bmod 33 = 17$

Q.2

a)) IDS are essential component in modern network security. Their primary purpose is to monitor n/w traffic & system for malicious activities, suspicious behavior or policy violation.

The need for IDS includes:

1. Detection of known & unknown threats
2. Real time monitoring
3. Compliance requirement
4. Post attack forensics
5. Minimizing damage

Types of IDS:

1. NIDS: NIDS monitors n/w traffic for suspicious activity & potential attacks. It examines the data packet passing thru the n/w looking for unusual patterns & known attack signatures. NIDS typically deployed at strategic points within the n/w. NIDS is ideal for large n/w where monitoring individual host is impractical
2. HIDS: HIDS is installed on individual host. It focuses on detecting threat that directly target a specific machine. It analyzes os activity, system logs

It can detect malware, file modification
It is best for environment where
Critical systems require Protection
against Unauthorized access or intern
threats

Types of IDS on detection -

1. Signature based IDS
2. Anamoly based IDS
3. Hybrid IDS

b)) AES is a widely used symmetric encryption algorithm that is recognized for its security, speed & simplicity.

It is used for securing sensitive data in various application including govern financial & Commercial sectors

Process :-

a) 10 rounds for 128-bit

b) 12 " " 192-bit

c) 14 " " 256-bit

Steps involved :-

1. Key expansion (Key Schedule)

2. Initial round

3. Main Rounds

3a. Subbytes

3b. Shift Rows

3c. Mix Columns

3d. Add Round Keys

4. Final Round

Summary

AES is industry standard for symmetric encryption offers a combine of strong security, efficiency & flexibility. Its resistance to cryptanalysis combined with high performance make AES a fundamental building block in modern cryptographic system